

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

Submit question no. 15 within first fifteen minutes.

1. What factors affects the size of digital representation of speech? What are some audio file formats? 3 + 2
2. Why is word recognition in natural language processing not a trivial task? 3
3. Why is smoothing of n-gram frequencies required? Why is the term *discounting* used in this context? 2 + 2
4. (a) What is a PoS tagset? Why are there multiple PoS tagsets for a language? 2 + 2
(b) Perform PoS tagging for the following sentence- 3
This is a very beautiful way to greet them.
5. Give examples of two different forms of verb phrases. What is *subcategorization* of verbs? 3 + 2
6. Give examples of ambiguities of different levels (viz., morphology, syntax, semantics, etc.) in English. How is Earley's parsing technique effective regarding ambiguity? 2 + 2
7. How can *unification of features* be incorporated in parsing using context free grammar? 3

8. How is *probabilistic context free grammar* useful? 2
9. Represent the following sentence using First Order Predicate Calculus- 2

Ram spoke to Sita and he got angry.

10. Identify lexical semantic relationships among the following words- 3
- glass (material), live (verb), sleep, live (adjective), slip, have (auxiliary), glass (tumbler), have (eat), have (own), utensils, leave.
11. How can selectional restrictions help in word sense disambiguation? 3
12. How does *recency* influence resolving references. 2
13. What is *grounding* in dialogue? 2
14. Why is translation of weather reports considered easy in comparison to other translation tasks? 2

Roll No. _____

Detach this page and submit it within first fifteen minutes.

15. (I) Fill in the blanks 1 × 7
- (a) _____ (UTF-16 / UTF-32) is a variable length encoding.
- (b) The leaf symbols of a parse tree are _____ (start-symbol/terminal/non-terminal) of a context free grammar.
- (c) Application of λ – *reduction* on the expression $\lambda(x)P(x)(A)$ results _____.
- (d) *Would* is a _____ (homonym/homophone/synonym) of *wood*.
- (e) In information retrieval, retrieving all the relevant documents present implies highest _____ (precision/recall).
- (f) “Before he bought the coat, Ram compared the prices in several shops”. In this sentence, *he* is a _____ (cataphora/anaphora).
- (g) The *fluency* of sentences produced via machine translation, can be assessed using _____ (n-grams/parallel corpora).

(II) State whether each of the following statements are true or false- 1×6

- (a) Unicode is a 16-bit encoding scheme.
- (b) Witten-bell discounting is a smoothing technique.
- (c) Synonymy may lower the precision of information retrieval.
- (d) In the sentence “Kamal said that Vimal bought himself a new shirt” *himself* refers to Kamal.
- (e) *Disfluency* is a phenomenon in speech.
- (f) When translation is required between one another among several languages interlingua based translation is more appropriate approach.